

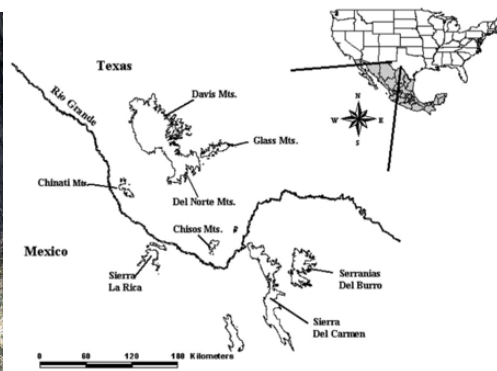
Lab, Week 10
3 November 2025

Pick a model, any model...

... to address the questions below.

Provide your answers to the questions below in a knitted RMarkdown file.

1. *You have been placed on an advisory board tasked with analyzing data on wildlife that could be affected by expansion of the border wall. Mexican black bears (*Ursus americanus eremicus*) inhabit a network of montane woodlands in Big Bend National Park, Texas, and into the Chihuahuan Desert of Mexico. Because they can only inhabit the mountains above ~1000 meters in elevation, they effectively exist as a metapopulation among mountain islands in a sea of desert. Dispersal among populations could be disrupted by a border wall, and the remaining populations in both the US and Mexico may be too small to persist. Staff at Big Bend National Park monitor the bear populations by keeping records on the number of reported sightings of black bears, especially females with cubs. The table below shows the number of sightings of black bears among three montane islands. (Data modified from Hellgren et al. 2005, *Biological Conservation* 122: 131-140.)*



Year	Mountain range	Sightings total	Females with cubs
1996	Davis Mts	26	15
1996	Glass Mts	32	12
1996	Chinati Mts	23	13
1998	Davis Mts	17	7
1998	Glass Mts	21	9
1998	Chinati Mts	31	14
2000	Davis Mts	28	12

- a. Find the MLE proportion of sightings of female bears with cubs by mountain range. Report these rates and their confidence intervals. Plot the results.

2000	Glass Mts	16	8
2000	Chinati Mts	13	5
2002	Davis Mts	5	2
2002	Glass Mts	8	2
2002	Chinati Mts	3	1

- b. Does the number of bears with cubs differs by mountain range and has there been a linear trend in numbers over time (i.e., include year as a *numeric* predictor variable)? Report your conclusion supported by statistics.

2. A border wall would separate the already dwindling northern and southern populations of the Mexican gray wolf, *Canis lupus* subsp. *baileyi*. **There are less than 250 Mexican gray wolves remaining in the wild** (though this number has been increasing with reintroduction efforts). These populations rely on relatively large ranges across which to search for prey, which include elk, cattle, deer, squirrels, rabbits, and other rodents. Ranchers often blame wolves for destroying cattle, but a surprisingly small proportion of their diet consists of cattle, with the majority being elk. The table below shows the number of scat (poop!) samples for which elk versus other prey items were predominant among three wolf packs in Arizona and New Mexico from 2005 to 2006. (Data modified from Merkle et al. 2009, *The Southwestern Naturalist* 54: 480-524.)



Year	Pack ID	Elk	Other (cattle, rodents)
2005	Blue Stem	28	5
2005	Saddle	28	2
2005	Luna	16	20
2006	Blue Stem	43	2
2006	Saddle	27	2
2006	Luna	38	12

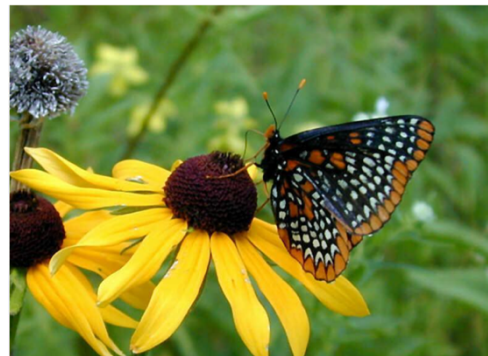
You have been placed on a public outreach task force to provide local ranchers with information on the actual impact of the wolves on their livestock.

Based on the limited dataset above, what proportion of scat samples did *not* consist predominantly of elk?

3. You've received a contract with the CDC to assess the potential for Lyme disease to be spread by avian hosts of ticks that often carry Lyme disease. Prior to collecting your own data, you've decided to look at previously published datasets to help design your study. Among these, Newman et al. studied the role of birds in northern California as hosts for *Ixodes pacificus* to try to understand if these birds might be potential reservoirs of Lyme disease (2015, PLoS ONE 10(2):e0118146). The file "LymeBirds.csv" contains the number of ticks removed from 623 birds, representing 53 species and 24 families.



- a. What is the mean number of ticks found per bird (and confidence intervals around this mean)? To answer this question, compare at least three models that you think could potentially describe the distribution of ticks found on birds in the dataset.
- b. Using the best model from a above, evaluate if there is an effect of the main habitat in which birds are found (MNHAB) on the number of ticks and use statistics to defend your results.
- c. Is there an effect of habitat or species on whether or not individual birds test positive for Lyme disease? Support your conclusion with statistics.
4. Flowering plant species contain different amounts of nectar per flower, and different concentrations of sugar in the nectar, which can affect the survival and reproduction of insects that feed on these flowers. You've been asked to help Shenandoah National Park choose pollinator-friendly plant species that will have the greatest chance of supporting local butterflies and other pollinators. To get started on this task, take a look at data collected Tufts REU students in 2018 on the sugar concentration of flowers of several species. Sugar concentrations have been log-transformed for easier analysis [using the formula $\log(\text{conc} + 0.5)$]. (Data from me.)



- a. What is the mean log-sugar concentration per species? Which two species have the highest concentration of sugar per flower? (Note: It may be helpful to back-transform the mean estimates and confidence intervals).
- b. Is there a difference between native and non-native species in sugar concentration per flower? Support your answer with statistics and provide means and confidence intervals for both.